

Enhancing Turbulence Resilience in Free-Space Optical Communications Using Germanium-Silicon Photodiode Array Interfaced with Multi-Core Fiber via Vertical Grating Couplers

Yuan-Zeng Lin, Yu-Han Lin, Cheng-Si Weng, Hsiang-Yi Kuo, Shin-Yu Lee, Chi-Wai Chow^(*)

Department of Photonics, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan,
cwchow@nycu.edu.tw

Abstract *We demonstrate a free-space-optical-communication (FSOC) with enhanced atmospheric-turbulence resilience using germanium-silicon (GeSi) photodiode-array interfaced with multi-core-fiber (MCF) via vertical-grating-couplers (VGC). Experimental results show the proposed scheme significantly improves FSOC performance under different turbulence conditions. ©2026 The Author(s)*

Introduction

Free-space optical communication (FSOC) [1, 2] has emerged as a promising technology for high-speed connectivity, offering advantages of inherent security, immunity to electromagnetic interference (EMI), and abundant unlicensed spectrum. However, practical FSOC deployment faces several challenges, including beam divergence over long distance transmission and atmospheric turbulence [3]. Among them, atmospheric turbulence is a critical issue, as it introduces random phase and amplitude fluctuations, leading to scintillation, beam wandering, and beam spreading. Adaptive optics (AO) [4] can be employed to compensate for turbulence induced distortions; however, such systems are often limited by response speed and bulky implementations. Alternatively, orbital angular momentum (OAM) [5] and mode division multiplexing (MDM) [6] techniques can enhance turbulence resilience; however, they typically involve complex generation and detection schemes. Furthermore, turbulence mitigation can also be achieved using integrated photonic processor [7] or multiple-input-multiple-output (MIMO) processor [8]; however, they may incur latency due to the training process.

To mitigate the optical power loss caused by beam divergence, a large-aperture collimator at the receiver (Rx) is desirable to collect more light without truncating the incoming optical beam. However, it could be bulky and costly. As an alternative, multiple Rxs with smaller apertures can be employed to replace a single large-aperture Rx. This raises the question of whether a multiple-aperture Rx system can achieve performance comparable to that of a large-aperture Rx. Previous study [9] has investigated multiple-aperture Rx system, where smaller apertures are used, and received signals are combined to enhance the signal-to-noise-ratio (SNR), approaching the performance

of a large-aperture collimator. Nevertheless, the hardware overhead, particularly the number of discrete photodiodes (PDs) increases significantly as the number of apertures increase. Fortunately, recent advances in silicon photonic (SiPh) fabrication enable the monolithic integration of a large number of devices, such as PDs, onto a single chip; hence, significantly reducing system complexity, cost and footprint.

In this work, we propose and experimentally demonstrate an integrated, turbulence-resilient FSOC scheme using a SiPh germanium-silicon (GeSi) PD array, interfaced with a multi-core fiber (MCF) via a vertical grating coupler (VGC) array. Experimental results show that a four-aperture Rx system processed with maximal-ratio combining (MRC) algorithm significantly outperforms a single Rx under weak, moderate, and strong turbulence conditions. A data rate of 39 Gbit/s is achieved, with BER below the 3.8×10^{-3} hard-decision forward error correction (HD-FEC) threshold even under strong turbulence.

SiPh Device and Maximal Ratio Combining (MRC) Algorithm

Fig. 1(a) illustrates the schematic of the proposed GeSi PD array interfaced with a MCF via a VGC array integrated on a SiPh platform. The chip was fabricated by IMEC on a silicon-on-insulator (SOI) substrate with a 220-nm top silicon layer and a 2- μm buried oxide (BOX) layer. Conventional grating couplers (GCs) typically require the optical fiber to be tilted at a small angle to suppress back-reflection induced by second-order Bragg diffraction. Here, the proposed VGC is specifically optimized to minimize Bragg reflection under normal incidence. The VGC is implemented using 193-nm deep-ultraviolet (DUV) lithography, with a 160-nm thick poly-silicon overlay to enhance grating directionality [10]. To ensure optimal coupling efficiency, the VGC geometry is

precisely arranged to match the core pitch of a 7-core MCF, which has a core diameter and core spacing of 8 μm and 41.5 μm , respectively. The whole MCF has cladding and outer coating diameters of 150 μm and 250 μm , respectively. The cross section of the MCF is shown in the inset. As shown in the microscopic image in Fig. 1(b), each VGC is connected to a high-speed GeSi PD featuring a lateral PIN structured with a 300-nm thick Ge absorption layer, enabling fast optical-to-electrical (OE) response. At a DC-bias of -3 V, the measured 3-dB bandwidth of the PD is > 50 GHz, with responsivity of approximately 1 A/W. This monolithic integration eliminates the need for multiple discrete commercial PDs and repeated fiber-to-chip alignments; hence, significantly reducing system footprint, complexity and cost.

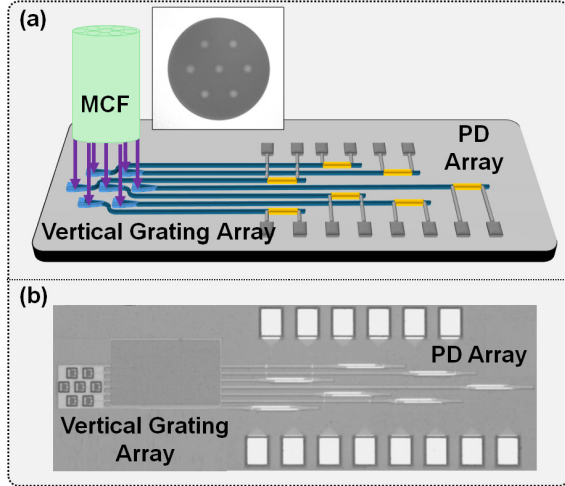


Fig. 1: (a) schematic of the proposed GeSi PD array interfaced with a MCF via a VGC array integrated on a SiPh platform. (b) Microscopic image of the fabricated chip.

A MRC algorithm is employed to combine the received signals. In FSOC systems, atmospheric turbulence induces independent fading across spatially separated apertures. Accordingly, in MRC, the optical signal received by each independent aperture i , denoted as $r_i(t)$, is weighted by a coefficient w_i proportional to its corresponding SNR. The combined output $y(t)$ can be expressed in Eq. (1),

$$y(t) = \sum_{i=1}^N w_i r_i(t) \quad (1)$$

where $w_i = \overline{H}_i / \sigma_{N_i}^2$ represents the combining coefficient, $\overline{H}_i$ is the channel response of the i -th channel, and $\sigma_{N_i}^2$ is the noise variance. This approach ensures that apertures with higher signal quality contribute more to the combined signal, effectively mitigating the impact of fading on individual apertures and maximizing the total

SNR of the link.

Proof-of-Concept Experiment

Fig. 2(a) shows the proof-of-concept experimental setup for evaluating turbulence resistant FSOC system using the GeSi PD array interfaced with a MCF via a VGC array. At the transmitter (Tx), a 1550 nm distributed feedback laser (DFB) diode is modulated by a Mach-Zehnder modulator (MZM) driven by an arbitrary waveform generator (AWG, Tektronix® 70001). An OFDM modulation is employed, including serial-to-parallel (S/P) conversion, quadrature amplitude modulation (QAM) symbol mapping, inverse fast Fourier transform (IFFT), cyclic prefix (CP) insertion, and parallel-to-serial (P/S) conversion. The FFT size is 512 with a CP length of 16, and adaptive bit-loading is applied. The modulated signal is amplified by an erbium-doped fiber amplifier (EDFA) and launched into free space through a large diameter collimator (Thorlabs® C80FC-C) with a 14.5 mm aperture. To simulate realistic atmospheric conditions, a SLM (Santec® SLM-200) is incorporated as an atmospheric turbulence emulator. By applying dynamic phase screens, weak, moderate, and strong turbulence conditions are generated. The SLM has a resolution of 1920 × 1200 pixels, a 7.8 μm pixel size, an 8 μm pixel pitch, and 10-bit grayscale depth. Then, the beam is expanded by an optical beam expander (Thorlabs® GBE10-C) to emulate the beam divergence over long-distance transmission. At the Rx, the optical light is collected by an array of collimators (Thorlabs® F810FC-1550) with 7

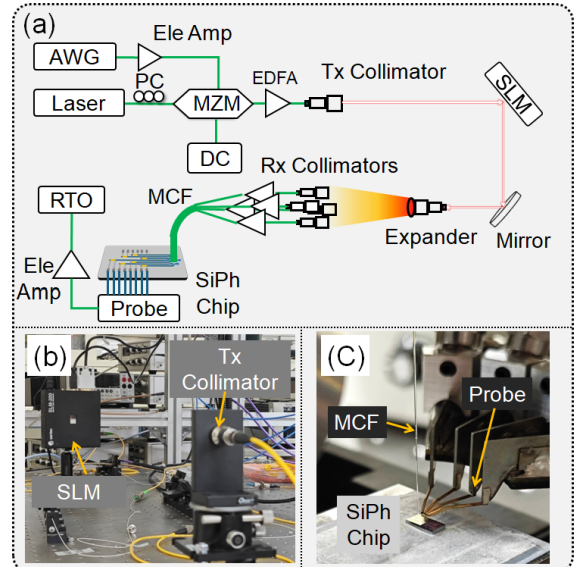


Fig. 2: (a) Proof-of-concept experimental setup for evaluating turbulence resistant FSOC system using the GeSi PD array interfaced with a MCF via a VGC array. Photos of (b) Tx collimator and SLM-based turbulence emulator; (c) SiPh chip interfaced with a MCF and RF probes.

mm aperture diameters. The collected optical signals are coupled into the MCF and coupled to the SiPh chip as shown in Fig. 1. The electrical signals after OE conversion by the PDs array are captured via RF probes connected to a real-time oscilloscope (RTO) (Tektronix® DPO71604DX). Finally, digital signal processing (DSP) is performed, including resampling, synchronization, CP removal, S/P conversion, FFT, symbol de-mapping, and MRC implementation. Although the MCF consists of 7 cores, only 4 cores are used to interface with 4 PDs, due to the limited number of input channels available on our RTO. Figs. 2(b) and (c) show the photos of Tx collimator with SLM-based turbulence emulator; and SiPh chip interfaced with MCF, respectively.

Results and Discussion

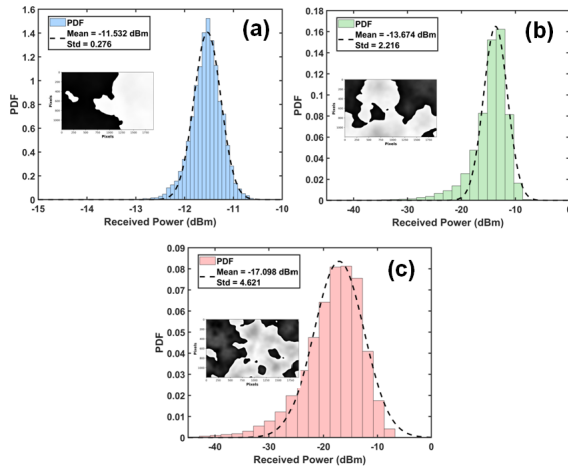


Fig. 3: Optical power PDFs under turbulence conditions of (a) weak, (b) moderate and (c) strong. Insets show the corresponding CGH phase patterns.

The statistical impact of atmospheric turbulence is characterized by the probability density function (PDF) of the received optical power, as shown in Fig. 3. The SLM is controlled by computer-generated hologram (CGH) patterns produced using the power spectrum inversion approach. The power spectrum follows the commonly used modified von Kármán spectrum [11], and the phase variation patterns are randomly and continuously generated by applying an IFFT to the corresponding spatial frequency power spectrum. The photos inseted illustrate the generated CGH patterns, where the vertical and horizontal axes represent the pixels on the SLM, and grayscale values from 0 to 1023 correspond to phases ranging from 0 to 2π . The refractive index structure parameters C_n^2 are set to 1.0×10^{-16} , 3.0×10^{-15} , and $1.7 \times 10^{-14} \text{ m}^{-2/3}$, representing typical weak, moderate, and strong turbulence scenarios for a simulated transmission distance of 500 km, respectively.

Under the weak turbulence conditions, as shown in Fig. 3(a), the power distribution is narrow ($\sigma = 0.276$), indicating a stable link. However, as the turbulence intensity increases to moderate and strong levels, as shown in Figs. 3(b) and (c), the PDF broadens significantly, with the standard deviation rising to 4.62, confirming the presence of intense scintillation.

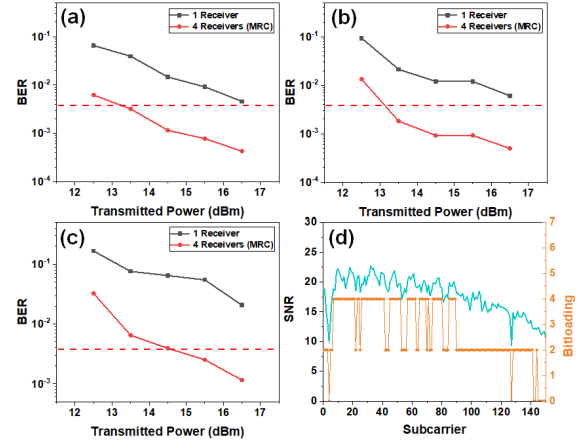


Fig. 4: BER curves comparing single- and four-aperture Rx with MRC under (a) weak, (b) moderate, (c) strong turbulence conditions. (d) SNR and bit-loading under strong turbulence condition.

Figs. 4(a)-(c) show the BER performance analysis for both single-aperture and four-aperture Rx processed with MRC algorithm. The four-aperture Rx provides significant diversity gain in all turbulence scenarios, and a data rate of 39 Gbit/s is achieved with BER below the 3.8×10^{-3} HD-FEC threshold even under strong turbulence. It can be observed that the single-aperture Rx cannot satisfy HD-FEC even under weak turbulence. Fig. 4(d) shows the measured SNR and bit-loading under strong turbulence using 4 Rx with MRC averaged by 250 times. The result indicates a relatively high bit-loading level of 4 (i.e. 16 QAM) can still be realized by the proposed scheme.

Conclusions

We proposed and proof-of-concept demonstrated an atmospheric turbulence resilient FSO system using GeSi PD array interfaced with a MCF via VGC array. The chip was fabricated on a standard SOI platform, and the VGC is specifically optimized to minimize Bragg reflection under normal incidence. The use of four-aperture Rx combined with MRC algorithm significantly outperforms the single-aperture case in all evaluated turbulence conditions. A 39 Gbit/s OFDM FSO transmission was achieved, satisfying the HD-FEC threshold even under strong turbulence of $C_n^2 = 1.7 \times 10^{-14} \text{ m}^{-2/3}$, demonstrating the robustness of the proposed scheme.

Acknowledgements

This work was supported by National Science and Technology Council, Taiwan (NSTC-113-2221-E-A49-055-MY3, NSTC-114-2218-E-011-003, NSTC-114-2640-E-A49-012, NSTC-113-2622-8-A49-013-SB).

References

- [1] E. Tangdiongga and T. Koonen, "Beam-steered optical wireless communication," *Proc. OFC 2025*, Paper W1H.1. DOI: [10.1364/OFC.2025.W1H.1](https://doi.org/10.1364/OFC.2025.W1H.1)
- [2] C. W. Chow, "Future perspectives on optical wireless communication – optical beam shaping, multiple access and integration (Tutorial)," *Proc. OFC 2025*, Paper W1H.4. DOI: [10.1364/OFC.2025.W1H.4](https://doi.org/10.1364/OFC.2025.W1H.4)
- [3] A. Trichili, M. A. Cox, B. S. Ooi, and M. S. Alouini, "Roadmap to free space optics," *J. Opt. Soc. Am. B*, vol. 37, pp. A184-A201, 2020. DOI: [10.1364/JOSAB.399168](https://doi.org/10.1364/JOSAB.399168)
- [4] S. Chen, S. Li, Y. Zhao, J. Liu, L. Zhu, A. Wang, J. Du, L. Shen, and J. Wang, "Demonstration of 20-Gbit/s high-speed Bessel beam encoding/decoding link with adaptive turbulence compensation," *Opt. Lett.*, vol. 41, pp. 4680-4683, 2016. DOI: [10.1364/OL.41.004680](https://doi.org/10.1364/OL.41.004680)
- [5] J. Wang, J. Y. Yang, I. M. Fazal, N. Ahmed, Y. Yan, H. Huang, Y. Ren, Y. Yue, S. Dolinar, M. Tur, and A. E. Willner, "Terabit free-space data transmission employing orbital angular momentum multiplexing," *Nature Photon.*, vol. 6, no. 7, pp. 488–496, 2012. DOI: [10.1038/nphoton.2012.138](https://doi.org/10.1038/nphoton.2012.138)
- [6] Y. Li, Z. Chen, Z. Hu, D. M. Benton, A. A. I. Ali, M. Patel, M. P. J. Lavery, and A. D. Ellis, "Enhanced atmospheric turbulence resiliency with successive interference cancellation DSP in mode division multiplexing free-space optical links," *J. Lightw. Technol.*, vol. 40, no. 24, pp. 7769-7778, 2022. DOI: [10.1109/JLT.2022.3209092](https://doi.org/10.1109/JLT.2022.3209092)
- [7] A. Martinez, A. Di Tria, F. Zanetto, F. Morandi, A. Milani, L. Resteghini, F. Morichetti, and A. Melloni, "Adaptive Bidirectional Free-Space-Optical Link Resilient to Atmospheric Turbulence," *Proc. ECOC 2025*, pp. 1-4, DOI: [10.1109/ECOC66593.2025.11262956](https://doi.org/10.1109/ECOC66593.2025.11262956)
- [8] C. W. Chow, Y. H. Jian, Y. H. Lin, C. W. Wu, W. L. Chen, Y. Z. Lin, T. Y. Hung, D. Yi, and H. K. Tsang, "Atmospheric turbulence resilient optical wireless communication (OWC) using silicon photonics (SiPh) programmable multiple input multiple output (MIMO) processor," *J. Lightwave Technol.*, vol. 44, pp. 1012-1019, 2026. DOI: [10.1109/JLT.2025.3606373](https://doi.org/10.1109/JLT.2025.3606373)
- [9] S. Y. Lee, Y. H. Jian, T. C. Wei and C. W. Chow, "Beam divergence and atmospheric turbulence resiliency enhancement in free space optical communication with multiple receivers and maximal-ratio combining algorithm," *Proc. OFC 2025*, paper Th2A.45.
- [10] Y. Tong, W. Zhou, and H. K. Tsang, "Efficient perfectly vertical grating coupler for multi-core fibers fabricated with 193-nm DUV lithography," *Opt. Lett.*, vol. 43, no. 23, pp. 5709–5712, 2018. DOI: [10.1364/OL.43.005709](https://doi.org/10.1364/OL.43.005709)
- [11] L. C. Andrews and R. L. Phillips, *Laser Beam Propagation Through Random Media*, 2nd ed. Bellingham, WA, USA: SPIE Press, 2005